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Enterprise Fax in 2025: A Strategic Assessment for Risk, Continuity and Transition

riot *¹

¹Institute for Modern Fax (IfMF)

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Executive summary

This position paper / conference talk transcript provides a strategic review of the role of fax technology in enterprises in 2025. We synthesize historical context, current usage patterns, security posture, regulatory drivers, and present a pragmatic transition roadmap. The goal is to equip executives, legal teams, and IT architects with concrete recommendations that balance compliance, operational continuity and modernization velocity.

*riot@c-base.org

1. Introduction

1.1. Background

Fax, historically known as telecopying or telefax, converts a scanned document to a bitmap, transmits it acoustically over PSTN or digitally over ISDN/VoIP gateways, and reconstitutes a paper or file at the receiver. The technology is simple, robust in band-limited channels, and critically easy to record on paper, which historically promoted its adoption in regulated sectors. See Wikipedia contributors [5] for a concise overview.

1.2. Motivation for this review

At many enterprise conferences the question persists: why is fax still used? This paper unpacks that question from the standpoints of compliance, operations, security and artful human factors (yes fax art is real and culturally interesting [4]).

2. Related Work

2.1. Prior analyses

Several retrospectives analyzed the stubborn persistence of fax (Broadcast [1]) and sector-specific studies document healthcare reliance (Smith and Müller [3]). Recent advisories highlight exploitable services (e.g., a high-profile advisory in 2024 related to a Windows fax service; see Center [2]) which we discuss in section 6.

3. Preliminaries and Definitions

3.1. Terminology

Metadata: information about the transmission (timestamp, caller ID).

Black Fax: a deneg attack using large black images to exhaust resources.

Fax Gateway: equipment or service translating between fax protocols (T.30, T.38) and email/file-based systems.

3.2. Assumptions

We assume a typical enterprise architecture: PBX and PSTN connectivity, one or more fax gateways (either on-premise or provided as a cloud service), and a document retention/legal discovery process.

4. Usage Patterns in Enterprises

4.1. Modes of operation

1. **Paper-centric:** Dedicated fax machines desktop-bound to phone lines.
2. **Hybrid:** Multi-function devices + server gateways (incoming faxes stored as PDFs, outgoing via SMTP bridge).
3. **Hosted:** Third-party cloud fax providers with number rental and archives.

4.2. Sector adoption

Healthcare and legal sectors show higher-than-average persistence due to regulatory requirements and cultural inertia.

5. Technical Anatomy: A Corporate View

5.1. Protocol summary

T.30 (analog fax) remains the baseline; T.38 (fax over IP) is used for VoIP scenarios. Gateways perform encoding/decoding and often provide logging and archival.

5.2. Enterprise integration

Enterprises commonly integrate fax with:

- DMS (Document Management Systems)
- Email servers (fax-to-email)
- Workflow engines (automated routing)

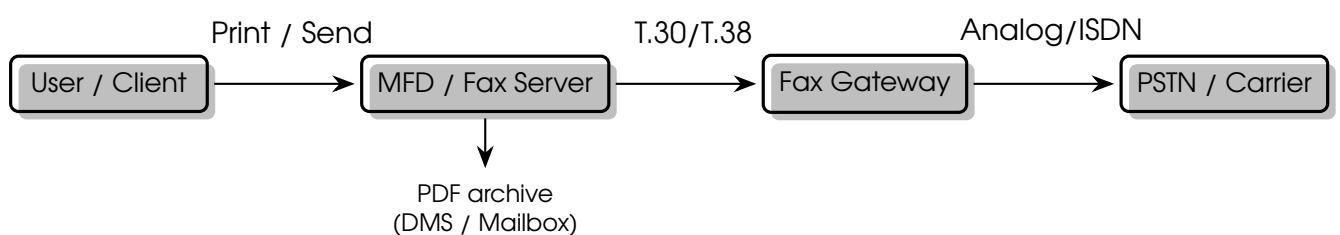


Figure 1: Canonical enterprise fax integration pattern.

6. Security Assessment

6.1. Threat model

Attack goals include data exfiltration, denial of service (DoS), persistence by exploiting fax-connected devices, and social engineering via trusted delivery channels.

6.2. Vulnerabilities and case studies

Historically, fax-connected devices (MFDs, fax servers) run complex stacks (imaging, embedded OS, network services). Remote code execution through document parsing and malformed transmissions is a live risksee the advisory cited in Center [2].

6.3. Attack surfaces

- PBX / SIP interface misconfigurations
- Fax-to-email pipelines lacking robust attachment scanning
- Legacy fax servers with unpatched OS components
- Physical security of paper archives

6.4. Mitigations (corporate best practices)

1. Isolate fax gateways on segmented networks; apply strict ACLs.
2. Restrict and log inbound/outbound numbers; apply rate limits.
3. Archive both the incoming TIFF/PDF and the transport metadata for chain-of-custody.
4. Eliminate on-device legacy services where possible; prefer hardened SaaS providers with strong SLAs and security attestations.
5. Regular vulnerability scanning and prompt patching.



Figure 2: High-level mitigation flow: isolate, detect, preserve.

7. Operational and Legal Considerations

7.1. Papertrail and admissibility

While fax produces tangible paper trails, admissibility varies by jurisdiction. Enterprises must harmonize retention policies with legal counsel and embed tamper-evident archival practices.

7.2. Business continuity

Fax remains an option for fallback communications when internet connectivity is lost; however, reliance solely on analog pathways is not a long-term resilience strategy.

8. Transition Roadmap: Mitigate, Monitor, Migrate

8.1. Phase 0 Inventory and Risk Triage

- Catalogue devices, gateways, phone numbers, and legal dependencies.
- Quantify monthly fax volume and categorize by workflow criticality.

8.2. Phase 1 Hardening

- Segmentation and patching; enable logging and archival.
- Deploy content inspection on fax-to-email paths.

8.3. Phase 2 Pilot Replacement

- Implement digital-native alternatives in low-risk workflows (secure DMS, signed PDFs).
- Run parallel operation and compare chain-of-custody outcomes.

8.4. Phase 3 Full Migration

Decommission devices where legally and operationally permissible; maintain limited legacy gateways for exceptional cases.

9. Cultural and Human Factors

A surprising driver of continued fax use is human trust and habit: faxes feel authoritative. Any migration plan must factor in training, legal sign-off and service-level parity otherwise users revert to fax.

10. Art, Humor and Corporate Messaging

A corporate presentation should balance seriousness and levity. Below is a short haiku (preserved for posterity) and a light corporate tagline.

10.1. Haiku (as presented at eloop)

Whirring paper sighs,
Ink-stained ghosts of old machines,
Time's echo lingers.

10.2. Corporate tagline

Enterprise Fax: minimizing operational disruption while accelerating secure, auditable digital transformation.

11. Recommendations

1. Establish executive-level ownership of fax de-risking (CISO + Legal).
2. Maintain a time-boxed hybrid stance: harden now, migrate where safe.
3. Invest in immutable archival and forensic-grade logging.
4. Engage vendors with SOC2/ISO27001 proof points when outsourcing fax services.
5. Run phishing and social engineering tests that include fax-based scenarios.

12. Conclusion

Fax in 2025 is a legacy transport that still matters rarely for technological superiority, often for compliance and habit. A careful, executive-backed, technically disciplined program will reduce risk and total cost of ownership while preserving continuity.

References

- [1] J. Broadcast. "Why Fax Refuses to Die." In: **Proceedings of RetroComms '03**. 2003.
- [2] Microsoft Security Response Center. **Windows Fax Service: CVE-2024-38104 Advisory (mock citation)**. Tech. rep. Microsoft, 2024.
- [3] A. Smith and B. Müller. "Fax in Health Care: Regulatory and Practical Considerations." In: **Journal of Health Tech** 12 (2019), pp. 45–59.
- [4] Various. **International Fax Art Exhibition Catalogue**. Archival material. 1996.
- [5] Wikipedia contributors. "Fax." In: (2023). <https://en.wikipedia.org/wiki/Fax>.

A. Supplementary Notes

A.1. Slide transcript and anecdotes

This appendix preserves some speaker notes and anecdotes from the original presentation materials:

- Anecdote: A lawyer recommended fax to receive a "transmission receipt" as a cheaper, faster alternative to registered mail.
- Observation: Thermal paper durability is low; prefer digital archival for longevity.

A.2. Example: Corporate policy template (excerpt)

Fax Usage Policy (excerpt)

Only approved numbers may receive PHI/PII. All inbound faxes with PHI must be stored in the corporate DMS with access controls and immutable retention for 7 years or as required by law.

A.3. Mock Appendix Figure: Trends (stylized)

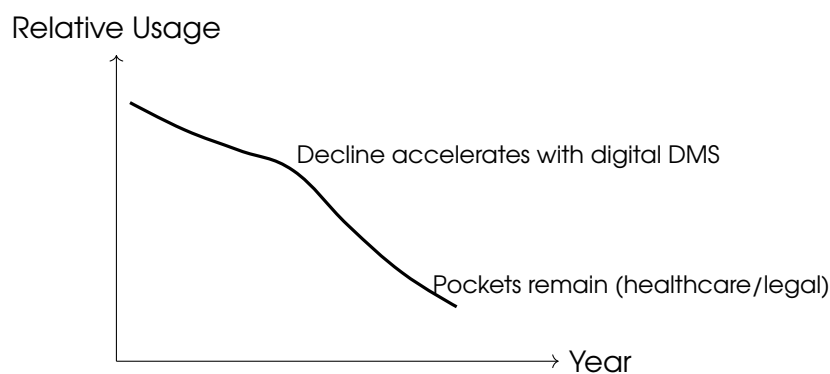


Figure 3: Stylized, non-quantitative trend of enterprise fax usage over time (illustrative).

Acknowledgements

Thanks to the community contributors who preserve institutional memory, and to the eccentric corners of the web where fax art lives.